

# A Relabeling Approach to Signal Patterns for Beacon-based Indoor Localization in Nursing Care Facility

Christina Garcia\*<sup>1</sup>, Vu Nguyen Phuong Quynh<sup>2</sup>, Haru Kaneko<sup>3</sup>, Sozo Inoue<sup>4</sup>  
<sup>1,2,3,4</sup> Graduate School of Life Science and Systems Engineering,  
Kyushu Institute of Technology, Kitakyushu 808-0135, Japan

## Abstract

In this study, we estimate the location of caregivers by investigating the signal patterns of detected beacons in different rooms and applying relabeling. Tracking the rooms visited by caregivers is useful to monitor patient and staff assistance. Estimating the visited rooms through the detection of beacons is affected by signal loss resulting in less data. In this study, we investigate the signal patterns in different rooms as the basis for relabeling to use the Received Signal Strength Indicator(RSSI) values from one location as the sample in another location to increase the training set and improve the model accuracy. RSSI-based features are extracted from augmented signal and Random Forest is used for location recognition. Beacon devices were installed in a nursing facility with a frequency setting of 10hz within 5 meters of coverage. The data was collected for 5 days. FonLog, a mobile application is installed in the phone carried by staff in the pocket which is used to gather RSSI readings and log location labels. Training data with increased samples from relabeling based on signal patterns of two rooms achieved an accuracy of 74% which is 5% higher than the performance of the original data.

keywords: relabeling, indoor localization, beacon, nursing care

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<sup>1</sup>alvarez7.christina@gmail.com

<sup>2</sup>quynh@sozolab.jp

<sup>3</sup>kaneko@sozolab.jp

<sup>4</sup>sozo@brain.kyutech.ac.jp

# 1 Introduction

Research focused on understanding the activity patterns of both patients and nursing staff to improve elderly care has continued to increase with the growing population of older adults. The aging population expands yearly, particularly in wealthy countries [17]. As a result, the utilization of nursing homes is also on the rise, which leads to an increasing patient-to-caregiver ratio. Considering the shortage of medical professionals, a balanced workload distribution is important to maximize staff and optimize care delivery [10]. To tackle this problem, different studies on staff-to-patient assistance and improving caregiver workload have emerged such as an automatic daily report generation system to record the activity of multiple residents in nursing homes [17].

Nurse care routine in facilities involves rounds to patient rooms and assisting the elderly in common areas. Different activities are commonly performed in respective stay areas in nursing homes [24] such as meal assistance in cafeteria, report creation in nurse station, and excretion assistance inside elderly rooms. Tracking nurse location and better comprehension of the care routine are also vital for developing patient assistance systems such as a nurse calling system which can assist patients with the nearest available nurse proposed to achieve a more time-saving response [13].

Studies on indoor positioning in hospitals target the location of patients, nurses [13], visitors, and equipment [5]. Conventional localization methods mainly focus on locating users equipped with wearable devices [1] but recently smartphones are preferred over wearable and are integrated to daily routines without causing any perceived disturbance [8]. With labeling and beacon data collected using installed application in smartphones, low data due to disabled location and Bluetooth settings can affect indoor localization performance.

Accuracy and reliability of the location information are crucial when used for ambient intelligence in the healthcare domain [4]. Addressing signal loss and improving both data quality and quantity are among the techniques to improve localization performance.

In this study, we investigate the signal patterns of detected beacons in various rooms and apply relabeling for the purpose of increasing training data to improve location recognition of low sample classes. Specifically, the objectives of this paper are:

- Real-world data collection deploying beacons in nursing care facility.
- Identification of signal pattern among different rooms as basis for increasing low sample classes.
- Propose a relabeling approach to signal patterns and compare indoor localization performance using original versus augmented data.

The subsequent parts of this paper are organized as follows: literature review on beacon technology and localization techniques in Section 2, overview of data collection in Section 3, methodology in Section 4 covering signal pattern and relabeling approach; results and discussion in Sections 5 and 6 with indoor localization performance, and Section 7 highlights the conclusion including future work.

## 2 Related Literature

This section provides an overview of the Bluetooth technology used in beacons, the application of indoor positioning in healthcare with localization techniques, and the utilization of data augmentation to address imbalanced real-world data challenges.

### 2.1 Bluetooth Low Energy

Bluetooth Low Energy (BLE) emitters used as beacons have shown improvement in technology enabling more people with different backgrounds to maximize BLE devices which are highly durable at reasonable prices [4]. As Bluetooth beacons perform better in coverage less than 4 meters [16], these emerging sensors are increasingly dispatched for positioning to overcome the ineffectiveness of the traditional navigation methods, for example, GPS, when measurements are obtained indoors. Other various applications that benefit from this technology are assisting blind people while traveling through unknown regions [7], ensuring the safety of individuals in indoor disaster situations [28], and supporting retailers to provide more sufficient customer location-based services [21].

### 2.2 Indoor Localization for Healthcare

In terms of digital technology utilization, healthcare is one of the fastest-expanding sectors. Japan projects a shortage of nearly one million medical workers by 2040 [3]. Indoor Positioning System (IPS) with technology such as Bluetooth allows monitoring of staff-to-patient assistance to optimize available care workforce and resources [25]. With beacons and the use of smartphones as gear, innovative healthcare applications include the development of a daily activity reminder module with a memory training set to assist patients with temporary memory loss to improve the quality of home care [11].

With continuously improving BLE technology, positioning solutions have emerged for application in healthcare but challenges in the on-site scenario and real-time localization has affected the accuracy of this technology [6]. Common challenges affecting beacon signal are susceptibility to diffraction, multipath propagation, angle-of-arrival, and absorption by human body [4].

Different approaches have been applied to achieve better accuracy for indoor localization from improving signal features with features-based proximity localization [4], using smoothing signal methods [24], to utilizing varying models such as model-based pedestrian dead reckoning [6], and machine learning for estimating stay areas of multiple residents [17]. To address the limitation of fingerprinting, indoor localization with adaptive signal representation has emerged recently based on spatial and temporal representations [15].

## 2.3 Data Augmentation

In hospital settings, it is challenging to collect annotated data resulting in limited and highly imbalanced datasets. To overcome this issue, one of the approaches applied in research is data augmentation. Data augmentation is a strategy used to enhance the size of a dataset owing to data scarcity [22]. However, because augmented data is meant to be indicative of real experimental data, it must be carefully created. Gaussian noise, local averaging, rotation, and temporal warping are some of the techniques used to create augmented data [20].

For indoor localization, annotating Received Signal Strength (RSS) data suffer from mislabeling and the uncontrolled real-world environment resulting in fewer beacon signals. With this, the data augmentation technique is usually conducted based on the RSSI values to improve the learning process [19, 18]. Currently, there are limited studies exploring the application of RSSI signal pattern techniques to increase training data for indoor localization in nurse care facilities.

## 2.4 Relabeling

Even though data augmentation is an efficient technique to enhance the performance of a network with limited data for training, it might cause the loss of label information, which can impair the performance of the model. Therefore, relabeling is applied to data augmentation of sensor data to preserve original label parameters [14]. Relabeling is also used to handle class imbalance by assigning new labels to minority classes [27].

The main goal of relabeling approach is to lessen the discrepancy in training data and improve model performance. In this paper, we utilize relabeling to update the label of augmented RSSI sample based on the signal pattern of another room.

For this study, beacons as BLE emitters are deployed in a nursing care facility to track staff-to-patient assistance. To address the challenge of limited RSSI signals in certain rooms, data augmentation techniques were employed. Specifically, signal patterns were observed from different rooms and were used as the basis for relabeling. While other studies use relabeling to preserve original label [14] or assign new labels to minority classes [27], relabeling on this research was utilized to assign a new room label to the augmented RSSI data from other rooms to match the minority class label.

## 3 Data Collection

The data was collected from the morning to afternoon in a span of 5 days and comprised of locations visited by a caregiver when performing the daily routine. Table 1 lists the nurse care tasks carried out by the caregiver. The majority of the activities performed were location dependent hence tracking the rooms visited by caregivers estimates the workload and patient assistance.

Table 1: Nurse care activities

| Activity Group                | Activity Name                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------|
| Patient care                  | bathing and wiping, changing, toilet/excretion assistance, daytime support,                    |
| Medical assistance            | meal preparation and assistance, recreation, checking patient vital, treatment, rehabilitation |
| Environmental arrangement     | cleaning and organization, linen change, washing                                               |
| Documentation / communication | special notes/communication, handwritten records,                                              |

### 3.1 Nurse Care Stay Area

FonLog, a mobile application is installed and used to gather RSSI readings and log location. The interface of FonLog in Fig. 1 is used for recording location labels and BLE ID is enabled to detect beacon data.

The caregiver was required to carry the smartphone while performing the nursing activities in different locations in one floor of the facility. The smartphone device is carried by the staff in the front pocket of the uniform to avoid disturbance during daily routine maintaining the quality of nurse care. Table 2 lists the locations visited by the participant during data collection including elderly rooms, common area, staff area, and hallway.

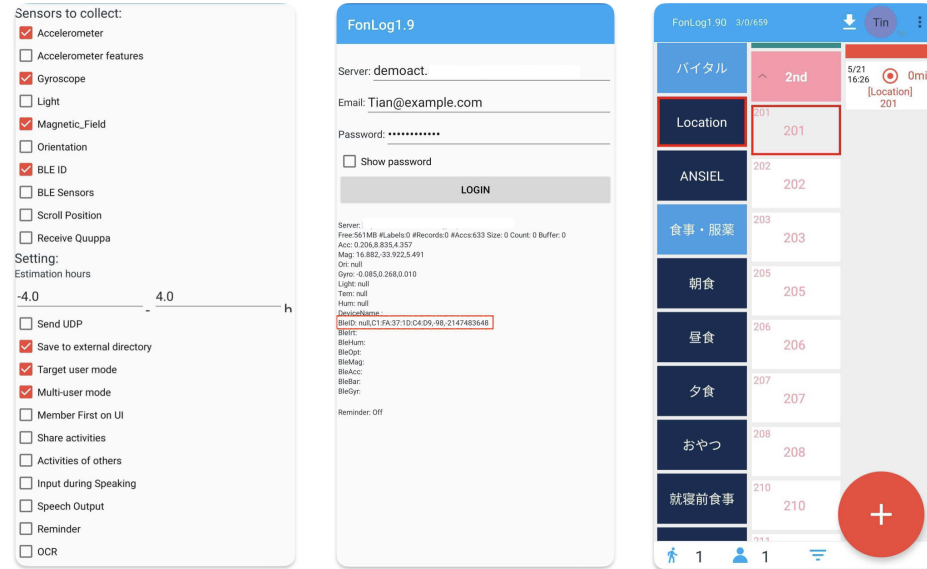


Figure 1: Mobile application Fonlog used in data collection

Table 2: Nurse care stay area

| Category      | Location      | Total Duration of Stay (mins) | Visitation Count |
|---------------|---------------|-------------------------------|------------------|
| Elderly Rooms | Room 01       | 13.117                        | 5 times          |
|               | Room 02       | 7.783                         | 7 times          |
|               | Room 03       | 22.549                        | 5 times          |
|               | Room 05       | 3.833                         | 4 times          |
|               | Room 06       | 11.716                        | 6 times          |
|               | Room 07       | 0.383                         | 2 times          |
|               | Room 08       | 9.466                         | 7 times          |
|               | Room 10       | 13.55                         | 6 times          |
|               | Room 11       | 8.649                         | 5 times          |
|               | Room 12       | 31.283                        | 6 times          |
|               | Room 13       | 58.35                         | 10 times         |
|               | Room 15       | 15.716                        | 4 times          |
|               | Room 16       | 0.483                         | 2 times          |
|               | Room 17       | 12.133                        | 6 times          |
|               | Room 18       | 4.25                          | 7 times          |
|               | Room 20       | 30.133                        | 10 times         |
|               | Room 21       | 1.416                         | 2 times          |
|               | Room 22       | 26.483                        | 8 times          |
|               | Room 23       | 294.466                       | 27 times         |
| Common Area   | Cafeteria     | 218.833                       | 190 times        |
|               | Kitchen       | 320.216                       | 116 times        |
|               | Cleaning      | 37.599                        | 38 times         |
| Staff Area    | Nurse Station | 299.316                       | 144 times        |
| Others        | Hallway       | 57.833                        | 63 times         |

The visitation count details the number of times a stay area is visited by the participant as recorded in FonLog as location label. Total duration is sum of all stay duration in visitation count. Both columns provide an overview of which location class has the least beacon data and which elderly room needed longer or more frequent assistance.

### 3.2 System Architecture

Initially, two separate smartphones are setup for the caregiver participant and labeler. The respective sensor data to be collected are enabled in FonLog including BLE ID. The location and bluetooth settings of the smartphone are also enabled and beacon detection is confirmed in the login interface where the mac addresses are displayed.

Matching RSSI from detected beacons and the corresponding location label can result to low class sample when the timestamps do not match which usually happens when the caregiver records the activity after performing different tasks. To address this issue, the labeler in the data

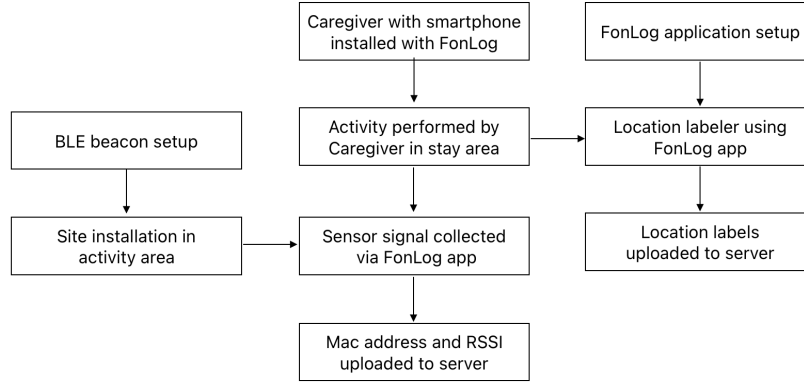


Figure 2: Overview of data gathering set-up

collection is an observant to the caregiver performing the daily routine. Fig. 2 details the system architecture of the data collection. The labeler chooses the location from the list of rooms in the app and records stay duration by clicking start and stop. Both location and labels are temporarily stored in the local storage of smartphones and uploaded to the same server when connected to the internet.

### 3.3 Area of Coverage

BLE beacons with a frequency setting of 10hz were installed on one floor of the facility detailed in Fig. 3. Each beacon was set to a detection level of zero up to five meters of coverage matching a transmission power (decibel-milliwatts) of up to -30dBm. Transmission power levels vary in calibrations of RSSI at 1 meter. Devices set in wider coverage have higher transmission power. Based on the specification sheet of the beacon used, the detection of RSSI at 100 meters has transmission power of +4dBm.

A total of 25 beacons were placed at a minimum height of 2 meters outside the patient rooms and in the common area covering the route of the caregiver. The placement of BLE beacons is subject to significant restrictions in medical and elderly facilities. In studies where a beacon tag is attached to the user, scanners are suitably deployed in corners and sections where geometry of location is considered to calculate position using methods without employing machine learning [23]. Specifically, in this study, the placement was decided in agreement with the staff and to detect the passage of caregivers in the common area of the facility including hallways. In consideration of the privacy of the patients and to avoid disturbance of the elderly during installation and device checking, beacons were installed on the door outside each patient room in suitable height matching pre-deployment experiment. In contrast to [23], FonLog application was carried by the nurses serving as scanners while beacon emitters are placed stationary. Also, Random Forest was utilized for indoor localization. For room estimation considering attenuation, RSSI values from

all rooms were considered in the feature extraction. Also, signal patterns were observed extending to nearby rooms around the target room as the basis for relabeling elaborated in Section 4.1 based on Fig. 5.

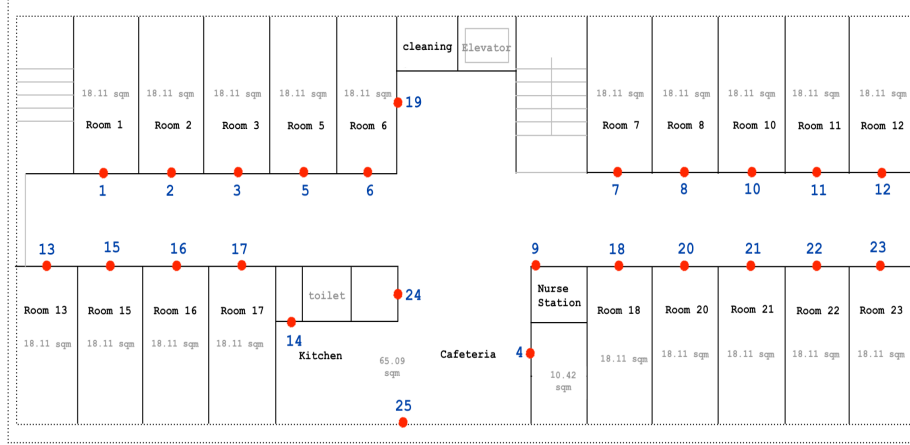


Figure 3: Floor plan of facility with installed beacons.

Respective mac addresses of the beacons were recorded prior deployment to match with the raw sensor data in Table 3 processed from the server. A unique User ID is assigned to each nurse as they log in FonLog. In the case of multiple users, User ID is used to identify respective nurses. Similarly, a unique customer ID is assigned to the patient attended. In this study, data was collected from only one participating nurse average in height. Any noticeable difference in height should be noted and the effect on the RSSI value should be observed in the case of multiple users.

Table 3: Collected beacon data from server

| User ID | Timestamp            | Mac Address       | RSSI |
|---------|----------------------|-------------------|------|
| 90      | 2023-04-13 T13:51:36 | EE:E7:46:DC:19:6F | -92  |
| 90      | 2023-04-13 T13:51:38 | 4B:A8:3B:03:6D:BF | -62  |

## 4 Methodology

The beacon data collected in Table 3 consists of User ID, timestamp, mac address, and RSSI. For localization, RSSI-based features are utilized. The timezone of the location label and beacon data were matched. From the raw location labels, entries lacking in start-time and stop-time with undefined duration were filtered. Duplicates and rows with null values were removed from the location label dataframe. Matching the mac address to the beacon list, the final RSSI dataframe is shown in Fig. 4.



## 4.1 Signal Patterns

| timestamp              | mac<br>address | RSSI |   | timestamp              | RSSI_1 | RSSI_2 | RSSI_3 | ... | RSSI_16 |
|------------------------|----------------|------|---|------------------------|--------|--------|--------|-----|---------|
| 2023-04-10<br>13:00:00 | 6              | -94  |   | 2023-04-10<br>13:00:00 | 0      | 0      | 0      | ... | 0       |
| 2023-04-10<br>13:00:00 | 6              | -96  |   | 2023-04-10<br>13:00:00 | 0      | 0      | 0      | ... | 0       |
| 2023-04-10<br>13:00:00 | 4              | -92  | → | 2023-04-10<br>13:00:00 | 0      | 0      | 0      | ... | 0       |
| 2023-04-10<br>13:00:00 | 6              | -94  |   | 2023-04-10<br>13:00:00 | 0      | 0      | 0      | ... | 0       |
| 2023-04-10<br>13:00:00 | 6              | -96  |   | 2023-04-10<br>13:00:00 | 0      | 0      | 0      | ... | 0       |

Figure 4: Matrix of received signal strength from each beacon

A total of 25 beacons and 19 elderly rooms were on the corresponding floor covered. Suppose  $RSSI_t$  is the detected RSSI at timestamp  $t$  at location  $m$ , we express RSSI measurement for all beacons  $n$  with the corresponding location label as

$$RSSI_t = [RSSI_{t1}, \dots, RSSI_{1n}], label_t = m_t \quad (1)$$

The signal database can be expressed as 2 and the labels as 3

$$RSSI = \{[RSSI_{11}, \dots, RSSI_{1n}], \dots, [RSSI_{t1}, \dots, RSSI_{tn}]\} \quad (2)$$

$$label = \{m_1, m_2, m_3, \dots, m_t\} \quad (3)$$

where  $t$  is the timestamp,  $n$  is the beacons in site from 1 to 25, and  $m$  is the corresponding elderly room. From the train data, location  $m$  with the least duration of stay based on Table 2 is identified.

The performance of localization decreases as the distance between the transmitter and receiver increases [26]. Considering the beacon setting within 5 meters coverage, signal patterns are limited to the corresponding six beacons in Fig. 5 which are investigated for each room. A list of beacons are created for each location such that  $f_m = [fl_m, sl_m, s_m, f_m, sr_m, fr_m]$  where  $s$  is the beacon on the source room,  $f$  is the beacon on the room in front of source,  $fl$  and  $sl$  are the beacons on the left side while  $sr$  and  $fr$  are beacons on the right side respectively. As an example in Fig. 5 with the source room as target location, the corresponding beacons are  $fl = 8$ ,  $sl = 20$ ,  $s = 21$ ,  $f = 10$ ,  $sr = 22$ , and  $fr = 11$  such that  $f_{21} = [8, 20, 21, 10, 22, 11]$ .

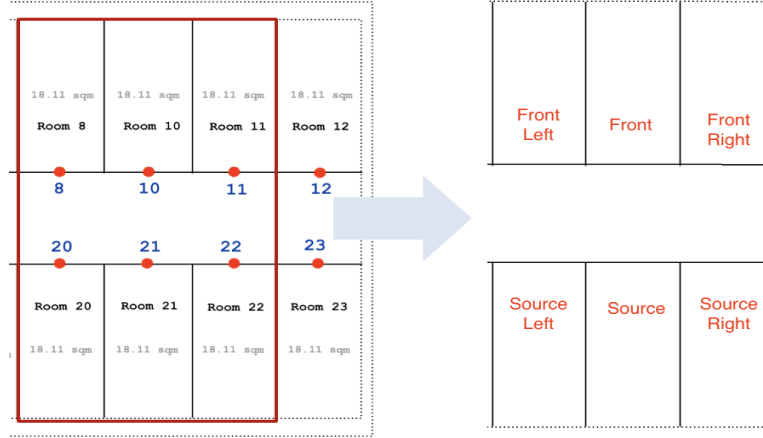


Figure 5: Six rooms with observed signal patterns

A matrix  $pattern_m$  containing the RSSI values of the beacons in the list  $f_m$  is created to be used for relabeling. For each beacon in  $f_m$ , statistical measures including standard deviation (STD), mean and signal count are calculated which signal patterns are based. To observe signal patterns, the data with the RSSI matrix in equation 2 and  $labels$  containing room numbers are required. The goal is to identify the locations with matching signal patterns  $pattern_{match}$  and  $pattern_{min}$  to proceed with relabeling. Enumerated in 1 to 4 are the steps plot signal patterns for all rooms.

1. Separate train data,  $d_{train}$  and test data,  $d_{test}$ .
2. Identify rooms with small sample in train data based on less than 1 minute duration of stay in Table 2 in section 3.1.
3. Filter 6 beacons for all rooms following Fig. 5.

$df = d_{train}$

for  $i$  in  $labels$ :

$d_i = df[df['room'] == "i"]$

if  $i == "1"$ :  $f_1 = [fl_1, sl_1, s_1, f_1, sr_1, fr_1]$

...

elif  $i == "23"$ :  $f_{23} = [fl_{23}, sl_{23}, s_{23}, f_{23}, sr_{23}, fr_{23}]$

for  $x$  in  $f_i$ :

$pattern_i = [d_i[x_0], d_i[x_1]...d_i[x_5]]$

$frames = \{\}$

4. Calculate statistical mean, count, and std for each of 6 beacons

for  $key$ , value in enumerate  $pattern_i$ :

$frames[key] = value$

$frames[key].count()$

$frames[key].std()$

$frames[key].mean()$

After plotting all the statistical values, we observe the trend of the signals in  $pattern_{min}$  and locate similarity in the rest of  $pattern_i$  from all other

rooms to find  $pattern_{match}$ . Initially, we filter the consideration to the rooms with the same geometry and room dimension based on the floor plan in Section 3.3. Next, the trend of the plot for signal count is matched such that the source and front rooms have higher count compared to the side rooms. Next, the trend for the mean is observed to get an estimate of the average RSSI value based on the central tendency of the data. Lastly, the deviation of data points from the mean indicating signal variability is inspected from the STD plots.

## 4.2 Relabeling Approach

For relabeling, train data  $d_{train}$ , the signal pattern  $pattern_{min}$  for the location with low sample, and the signal pattern  $pattern_{match}$  of the identified matching location are required. The expected output is the augmented train data  $d_{train,new}$  with concatenated original train data  $d_{train}$  and relabeled data  $d_{relabelled}$ . Enumerated in 1 to 5 are the steps applied to perform relabeling.

1. Create a dataframe  $d_{relabelled}$  with columns from  $pattern_{min}$ .
2. Map the values from  $pattern_{match}$  to  $d_{relabelled}$ .
3. In  $d_{relabelled}$ , create columns for beacons not included in  $pattern_{min}$  following 1 and fill with zero values.
4. For relabeling, assign label to  $d_{relabelled}$  with labels from location  $m_{min}$ .
5. Create a dataframe  $d_{train,new}$  fill with concatenated train data  $d_{train}$  and  $d_{relabelled}$ .

Nursing facilities and hospitals commonly adopt a rectangular layout with uniform space allocation within rooms. Considering the impact of environment on the signal, the relabeling approach is targeted to floors that consist of rooms with identical geometry and dimensions. In the original floor plan depicted in Fig. 3, the relabeling method is exclusively applied to patient rooms. Relabeling does not apply to signals from beacons installed in locations which exhibit varying dimensions and layouts such as open areas including the Kitchen and Cafeteria, as well as common rooms like the Cleaning and Nursing station. Similarly, the same beacon sensors should be installed in uniform height.

## 4.3 Indoor Localization

The timestamp of location labels and RSSI data are matched. Statistical and time-based features were extracted from the RSSI matrix. Eight statistical features were extracted specifically mean, variance, STD, minimum, maximum, sum, median, and count. Including the number of detected signals per device in the features resulted in better model performance [9]. The time-based features extracted are hour, minute, and microsecond. 45 seconds time window was applied with no overlapping. Following the suggested train-test ratio from empirical studies [2], the train and test data were separated by 70% to 30% split. Random Forest is used for location recognition as this classifier demonstrated effectiveness in handling imbalanced data [12]. The `train_test_split` function of

RF model was used with the random state hyperparameter set to a fixed integer value to preserve the split across different executions.

## 5 Results

Fig. 6 provides an estimate of the rooms visited by the nurse with the shortest duration of stay.

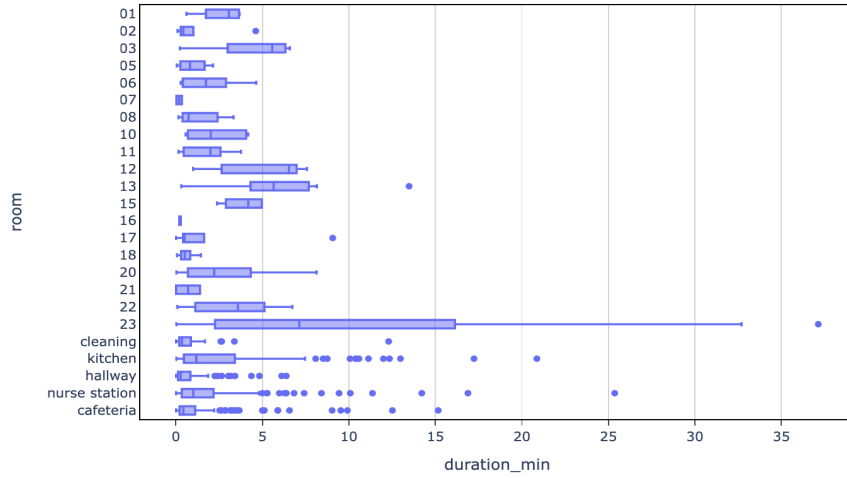


Figure 6: Duration of stay per location label

Based on the boxplot in Fig. 6 and Table 2 in section 3.1, we identified Room 16 and 7 as the classes which need more samples. We then proceed with checking the signal patterns of these rooms to find a match in  $pattern_i$ .

### 5.1 Received Signal Strength

From the signal patterns of the rooms with similar layout to Rooms 16, we identified Room 20 as a match. The signal patterns of both rooms are shown in Fig. 7 where FL, SL, S, F, SR and FR follow the abbreviations introduced in Section 4.1, Fig. 5. Examining the trend of the plot for signal count, both Room 16 and 20 detected more signal from beacons located in the source and front room. Also, it follows that *fl* has the least detected signal while *fr* has comparatively higher count than the rest of the beacons from the side rooms. Comparing the plots of the mean RSSI, both rooms display the highest signal value following the beacon with the highest count. With the STD plots, both Room 16 and 20 also display highest variability on the beacons with most RSSI detected. For Room 7, matching pattern was obtained from Room 1.

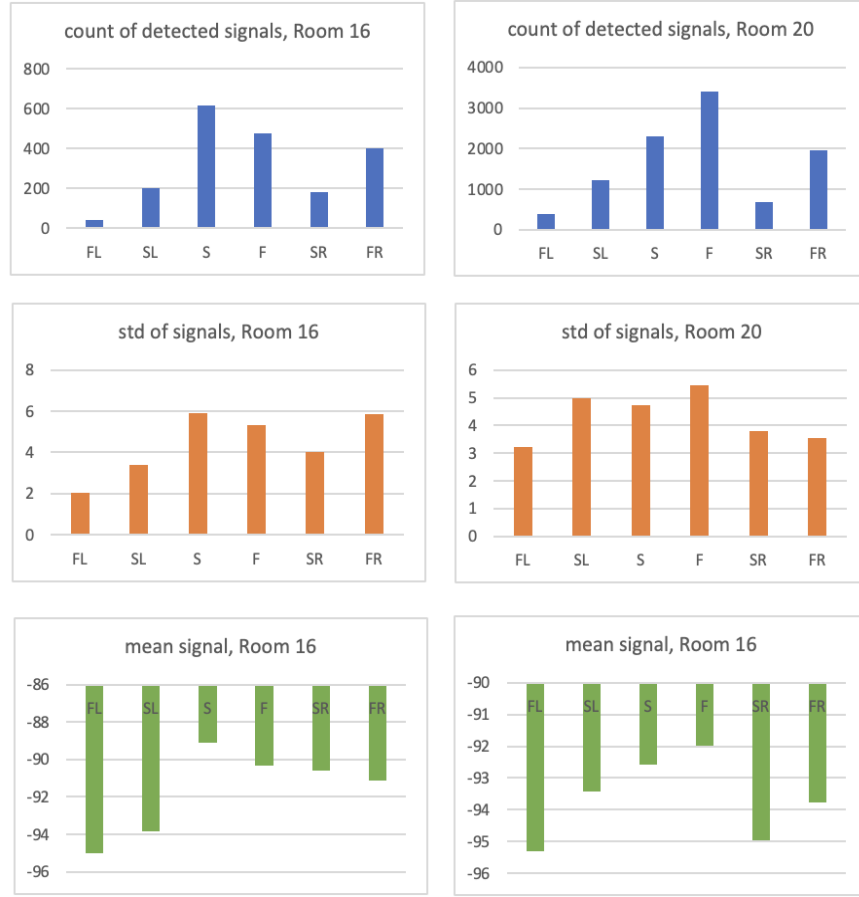


Figure 7: Signal pattern of Rooms 16 and 20

## 5.2 Indoor Localization

The indoor localization was approached as recognition using machine learning with rooms as the classes. Table 4 summarizes the comparison of indoor localization using the original and augmented data. Random Forest achieved higher performance in all metrics.

The original train data was increased 27% after combining the relabeled data. Both Precision and Recall showed a slight increase of 3% and 5% respectively. F1-score which displays a balanced trade-off between precision and recall has increased to 64% from the original 48%. The model achieved an accuracy of 74% when training with then augmented data which is slightly higher than the original 69%.

Examining the performance of Rooms 16 and 7 in Table 4, we were able to confirm that increasing the samples with relabeling resulted in better location recognition. From the confusion matrix in Fig. 8, both

Table 4: Performance Metric, Original versus Augmented Data

| Location     | Original Precision | New Precision | Original Recall | New Recall  | Original F1-score | New F1-score |
|--------------|--------------------|---------------|-----------------|-------------|-------------------|--------------|
| Room 3       | 1.00               | 1.00          | 0.50            | 0.50        | 0.67              | 0.67         |
| Room 6       | 1.00               | 1.00          | 1.00            | 1.00        | 1.00              | 1.00         |
| Room 7       | 0.00               | 1.00          | 0.00            | 1.00        | 0.00              | 1.00         |
| Room 8       | 1.00               | 1.00          | 0.33            | 0.33        | 0.50              | 0.50         |
| Room 10      | 0.00               | 0.00          | 0.00            | 0.00        | 0.00              | 0.00         |
| Room 11      | 0.80               | 0.80          | 0.80            | 0.80        | 0.80              | 0.80         |
| Room 12      | 0.64               | 0.64          | 0.78            | 0.78        | 0.70              | 0.70         |
| Room 13      | 0.67               | 0.73          | 0.89            | 0.89        | 0.76              | 0.80         |
| Room 15      | 1.00               | 1.00          | 0.33            | 0.50        | 0.50              | 0.67         |
| Room 16      | 0.00               | 0.50          | 0.00            | 1.00        | 0.00              | 0.67         |
| Room 17      | 1.00               | 1.00          | 0.75            | 0.75        | 0.86              | 0.86         |
| Room 18      | 0.00               | 0.00          | 0.00            | 0.00        | 0.00              | 0.00         |
| Room 20      | 0.73               | 0.73          | 0.89            | 0.89        | 0.80              | 0.80         |
| Room 22      | 0.25               | 0.33          | 0.50            | 0.50        | 0.33              | 0.40         |
| Room 23      | 0.67               | 0.62          | 0.80            | 0.80        | 0.73              | 0.70         |
| <b>Total</b> | <b>0.71</b>        | <b>0.74</b>   | <b>0.69</b>     | <b>0.74</b> | <b>0.48</b>       | <b>0.64</b>  |

Rooms 16 and 7 were misclassified training with the original data. Using the same test data in Fig. 9, correct classification was achieved for both Rooms 16 and 7 after relabeling.

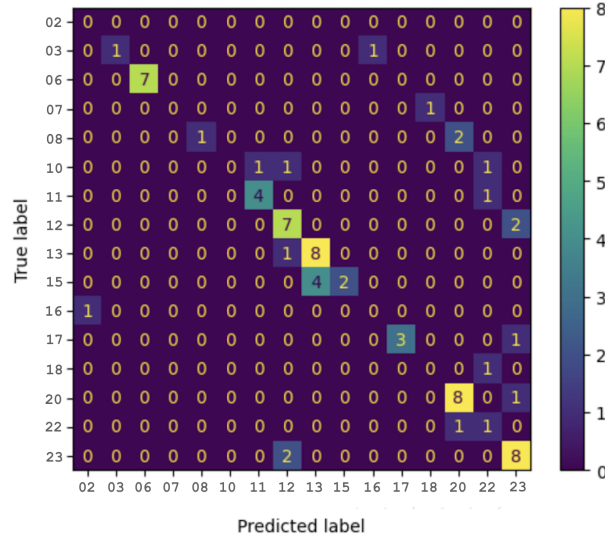


Figure 8: Confusion matrix, Original Data

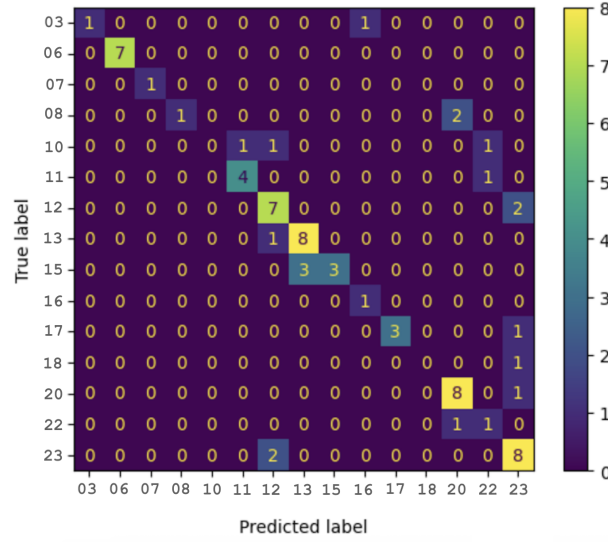


Figure 9: Confusion matrix, with Relabeled Data

## 6 Discussion

Comparing the performance of the model in Fig. 10, it is notable that increasing the training data with relabeling based on signal pattern resulted to improved location recognition. However, using the same test data to evaluate on different training set has displayed limited evaluation with fewer samples for the targeted Rooms.

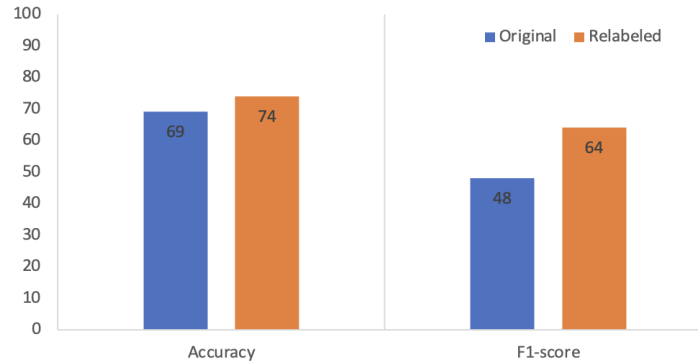


Figure 10: Comparison of Indoor Localization Performance

Considering the limited test data, we performed a 2-fold cross-validation for Room 16 using the training data to further elaborate the performance

where we concatenated the relabeled samples to the 50% of original training data and used the remaining 50% for validation. The result showed that performance for Room 16 has increased even with half of the training data with relabeled samples combined. On the other hand, the accuracy of other classes has decreased as expected with half of the training data utilized by the model as compared to the full training data used. This shows that the amount of training data affect model performance.

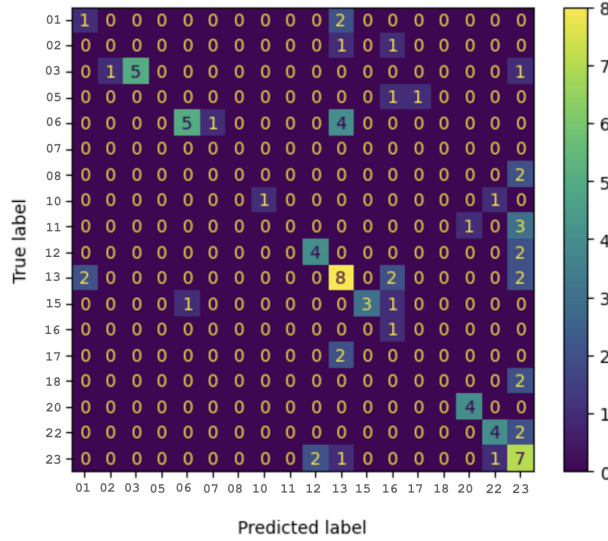


Figure 11: Confusion matrix, Cross Validation

The confusion matrix in Fig 11 show that Room 16 is still correctly classified during cross validation even with a decrease of 42% in training data. Table 5 compares the Precision, Recall and F1-score specifically for Room 16 training under different data.

Table 5: Location Recognition Comparision, Room 16

| Evaluation       | Precision | Recall | F1-score |
|------------------|-----------|--------|----------|
| Original data    | 0.00      | 0.00   | 0.00     |
| Augmented data   | 0.50      | 1.00   | 0.67     |
| Cross-validation | 0.17      | 1.00   | 0.29     |

## 7 Conclusion and Future Work

The challenge of data collection on-site in nursing care facility results to limited data which affects the performance of indoor localization. In this paper, we investigated the signal patterns in different rooms as the



basis for relabeling to increase the training data and improve location recognition. Training data with increased samples of Room 16 and Room 7 from relabeling according to signal patterns of Room 20 and 1 respectively achieved an accuracy of 74% which is 5% higher than the original 69%.

Though we have shown that relabeling can improve indoor localization, the quantity of the data collected is still limited. For our future work, we will continue data collection and evaluate the performance with more samples in the test data. Furthermore, we will improve the relabeling approach to cover locations in the common area without similar geometry and unequal dimension such as the cafeteria where nursing activities are performed longer. Also, we will delve deeper into the analysis of signal patterns by incorporating additional statistical analysis techniques. As for the model, we will try feature learning to extract more comprehensive features from the RSSI values and improve segmentation by setting window size depending on the frequency of the detected signals. To improve results for Room 16, we will explore utilizing synthetic approaches like SMOTE and ADASYN to increase sample size.

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